

All the Math You Missed

Chapter 10: Elementary Number Theory

Solutions

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Problem 1

Show that for any integer $k \geq 2$, the quantity $\sqrt[k]{p}$ is irrational for any prime number p .

Proof

Suppose $\sqrt[k]{p}$ is rational. Then there exist integers a and $b \neq 0$ such that

$$\sqrt[k]{p} = \frac{a}{b}.$$

Raising both sides to the k th power gives

$$p = \frac{a^k}{b^k},$$

so that

$$pb^k = a^k.$$

This implies that $p \mid a^k$. Since p is prime, it follows that $p \mid a$. Let $a = p \cdot m$ for some integer m .

Then,

$$a^k = (p \cdot m)^k = p^k m^k,$$

and so

$$pb^k = p^k m^k \implies b^k = p^{k-1} m^k.$$

This implies that $p \mid b^k$, so $p \mid b$ as well.

Therefore, both a and b are divisible by p , contradicting the assumption that $\frac{a}{b}$ is in lowest terms.

Hence, $\sqrt[k]{p}$ is irrational. ■

Problem 2

Show that $\sqrt{2} + \sqrt{3}$ is an algebraic number of degree 4.

Proof

To show that $\sqrt{2} + \sqrt{3}$ is an algebraic number, it suffices to show that it is a root of a polynomial with rational coefficients.

Let $x = \sqrt{2} + \sqrt{3}$. Then,

$$x^2 = (\sqrt{2} + \sqrt{3})^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}.$$

$$x^2 - 5 = 2\sqrt{6} \Rightarrow \left(\frac{x^2 - 5}{2} \right)^2 = 6.$$

So:

$$(x^2 - 5)^2 = 24 \Rightarrow x^4 - 10x^2 + 25 = 24 \Rightarrow x^4 - 10x^2 + 1 = 0.$$

Thus, $\sqrt{2} + \sqrt{3}$ is a root of the polynomial $x^4 - 10x^2 + 1$, and therefore is algebraic. ■

Problem 3

Find integers a and b such that

$$375a + 924b = 3$$

Solution

First, find the gcd(375,924). We'll use the Euclidean algorithm:

$$924 = 375(2) + 174$$

$$375 = 174(2) + 27 \quad ***$$

$$174 = 27(6) + 12 \quad **$$

$$27 = 12(2) + 3 \quad *$$

$$12 = 3(4) + 0$$

$$3 = 3$$

How convenient, this is the number we want for our initial equation. Now, we can use our process above to "unwrap" our solution. We'll begin by taking $*$ and going backwards.

$$3 = 27 - 12(2) \quad *$$

We can use $**$ to write 12 as $174 - 27(6)$. So, we now have:

$$3 = 27 - [174 - 27(6)](2)$$

$$3 = 27(13) - 174(2)$$

Likewise, we use $**$ to rewrite 27 in terms of 174 and $***$ to rewrite 174 in terms of 375 and 924:

$$3 = [375 - 174(2)](13) - [924 - 375(2)](2)$$

$$3 = [375(13) - 174(26)] - [924(2) - 375(4)]$$

$$3 = 375(17) - 174(26) - 924(2)$$

$$3 = 375(17) - [924 - 375(2)](26) - 924(2)$$

$$3 = 375(17) - 924(26) + 375(52) - 924(2)$$

$$3 = 375(69) - 924(28).$$

Thus, we have our solution: $375a + 924b = 3$ when $a = 69$ and $b = 28$.

Problem 4

Find the real number γ with the continued fraction expansion

$$\gamma = 1 + \frac{1}{1 + \frac{1}{1 + \dots}} = [1 : 1, 1, \dots].$$

Solution

We begin by noticing that our continued fraction contains itself. That is, the first denominator is actually the same value as our entire expression γ . Thus, we have the following:

$$\gamma = 1 + \frac{1}{\gamma}$$

Thus,

$$\gamma^2 - \gamma - 1 = 0.$$

This is a quadratic equation:

$$\frac{1 \pm \sqrt{5}}{2}.$$

The golden ratio!

Problem 5

For $a^2 + b^2 = 2c^2$, find eleven integer solutions.

Solution

We can find a collection of solutions by setting $a = b = c$. This simplifies our equation to

$$a^2 + a^2 = 2a^2$$

$$2a^2 = 2a^2$$

$$a = a.$$

So, any integer triple of the form (a, a, a) satisfies the equation.

- | | |
|------------|----------------|
| 1. (1,1,1) | 7. (7,7,7) |
| 2. (2,2,2) | 8. (8,8,8) |
| 3. (3,3,3) | 9. (9,9,9) |
| 4. (4,4,4) | 10. (10,10,10) |
| 5. (5,5,5) | 11. (11,11,11) |
| 6. (6,6,6) | |